

Gamma-Ray Attenuation and S-Process Neutron Capture for measuring material properties

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Abstract

In this experiment, values of two material properties are found. A Hyperpure Germanium (HPGe) Spectrometer is used together with a computer for data analysis and recording (with the MAESTRO software). Values of attenuation coefficients, μ , of Lead (Pb) and Aluminium (Al) are found for different Gamma-Ray energies. These values are shown to vary from $0.61 \pm 0.02 \text{ cm}^2/\text{g}$ to $0.015 \pm 0.09 \text{ cm}^2/\text{g}$ for energies between 244.79 keV to 1408.01 keV in Lead. The general dependence of μ on gamma-ray energy is shown. Aluminium data is found to be inconclusive. The half-lives of Copper (Cu) and Indium (In) are also determined. This is done by using the phenomenon of Slow process (S-Process) Neutron Capture. These half-lives are found as 3.20 ± 0.34 mins for Copper and 95.9 ± 48.3 mins for Indium. The nature of large uncertainties in all measurements are discussed.

1 Introduction

The primary aim of this experiment is to gain a deeper understanding of radioactive processes. Two forms of radiation are studied, namely Gamma-Ray emission and Neutron Activation. These effects are used to determine the total attenuation coefficients of materials and the half-life of materials respectively. Gamma Ray Spectrometry is used to find the attenuation coefficients of Lead (Pb) and Aluminium (Al). This is done by application of Beer's law which gives the intensity of a photon beam travelling through matter [4]

$$I = I_0 \exp(-\mu x) \quad (1)$$

where I is the transmitted intensity, I_0 is the incident beam intensity, and μ and x are the mass attenuation coefficient and mass thickness of the absorbing material respectively. The 'mass thickness' of a material is the product of its linear thickness and its density.[5] It is heavily influenced by the atomic number of the material's constituent atoms. It is expected that for higher energy gamma-rays, the value of μ will decrease by some exponential law as suggested by Eq(1). This is because it is easier for higher energy rays to penetrate the material.

The half-lives of Copper (Cu) and Indium (In) are determined by taking advantage of the Neutron Capture process. This process (specifically s-process neutron capture) is essential for stellar nucleosynthesis. It's responsible for creating roughly half of atomic nuclei in stars which are heavier than Iron (Fe56).[6] Using this process, radioactive isotopes are created. Subsequently, a HPGe detector is used to measure the decay gamma-radiation from these isotopes which are in turn used to extrapolate the half-life of the material using the equation[7]

$$\frac{dN}{dt} = \epsilon_{abs} A_0 \exp\left(-\frac{\ln(2)t}{t_{1/2}}\right) \quad (2)$$

where $\frac{dN}{dt}$ is the count rate of the sample, ϵ_{abs} is the absolute detection efficiency, A_0 is the activity of the sample and $t_{1/2}$ is the half-life of the material. To be precise, a graph of $\ln(dN/dt)$ against time is plotted for some high energy gamma-ray peaks. The gradient of these graphs is used to find the half-life with the following equation

$$t_{1/2} = -\frac{\ln(2)}{grad} \quad (3)$$

where $grad$ is the gradient of the graphs mentioned above.

2 Method

2.1 Measuring Attenuation Coefficients

Before beginning the measurements of sample counts, it was necessary to calibrate the energies of gamma-ray emission peaks using a Europium source. The source was first placed on the sample stand using a pair of tweezers. The stand was previously adjusted so that the source was 10cm away from the base where the HPGe detector head sat. The stand, together with the source, were then placed on the detector. Once this was done, the detector was set for an acquire time of 10 minutes using the MAESTRO software. This produced a spectrum for Eu152. (A diagram of the detector and other equipment can be seen on the next page.)

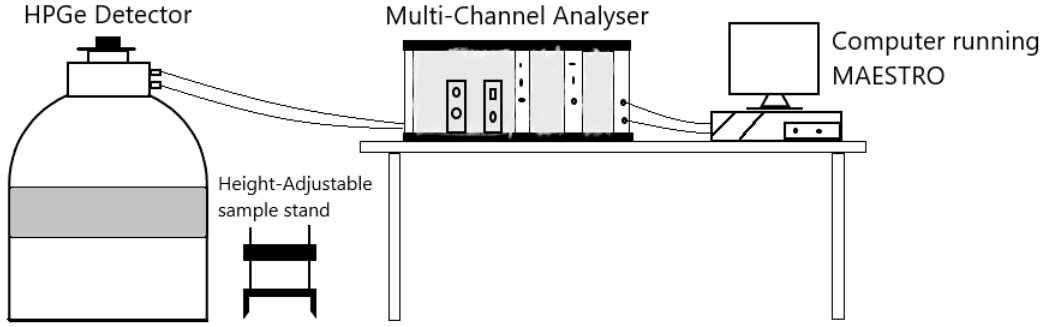


Figure 1: Diagram of equipment used: HPGe Spectrometer connected to a Multi-Channel Analyser and a computer. A sample stand can be seen on the ground which can be placed on top of the detector together with any small radioactive source.

The sample stand was then removed from the HPGe detector and a thin piece of Lead ($0.6g/cm^2$ mass thickness) was placed on its lower shelf below the source. The stand was subsequently placed back in the detector and monitored for 3 minutes: counts were recorded at 8 different gamma-ray energies. This process was repeated 5 times for different pieces of Lead of varying mass thicknesses ($0.6g/cm^2$, $1.22g/cm^2$, $1.42g/cm^2$, $1.8g/cm^2$ and $3.4g/cm^2$) followed by similarly varying Aluminium sheets ($6.75g/cm^2$, $16.5g/cm^2$, $19.6g/cm^2$ and $21.8g/cm^2$). The data recorded was used to calculate mass attenuation coefficients at these energies.

2.2 Measuring Half-Lives

Using a calibrated spectrum (just like in section 2.1), two regions of interest (ROIs) were set around the 121 keV and 1408 keV peaks (the highest and lowest energies on the spectrum). Once set, the full spectrum was saved and the ROIs were deleted. Using the calibrated spectrum, a new ROI was marked around 1039 keV and was saved as 'Copper'.

A non-irradiated copper sample was used to determine the background spectrum. The non-irradiated sample was placed on the detector for an acquire time of 3 minutes. After this, the background spectrum was saved and an irradiated copper sample was placed on the detector. As the irradiated sample would decay within 30 minutes, a stopwatch was used to keep track of time during the measurement to ensure that enough data points are recorded.

At 3 minute intervals, the gross number of counts and the error in the counts were recorded at the ROI around 1039 keV. The spectrum was then saved and cleared so that the next 3 minute measurement could start. This was repeated 5 times for the irradiated copper sample. Copper half-life was calculated by finding the best-fit values to the non-linear least squares problem. In this method, a sum of squares denoted χ^2 is minimised.

The copper sample was replaced with an irradiated Indium sample. An initial spectrum was taken which was then used to mark ROIs at 417 keV, 1097 keV and 1294 keV. After this, a .JOB file (INDIUM.JOB) was used to record the count at these peaks every five minutes for an hour. These readings were each saved in separate files. A graph of $\ln(\Delta N/\Delta t) - t$ was plotted for all 3 peaks. The gradients of these graphs were used with Eq(3) to calculate Indium's half life.

3 Experimental Results

Note: all gradient values and uncertainties in gradients were calculated using Excel's LINEST function unless stated otherwise.

A decaying exponential relationship was found between values of μ , the mass attenuation coefficients, and the gamma-ray energies between 244.79keV and 1408.01keV for Lead. This is consistent with what was predicted by theory.[1] The below graph displays this result.

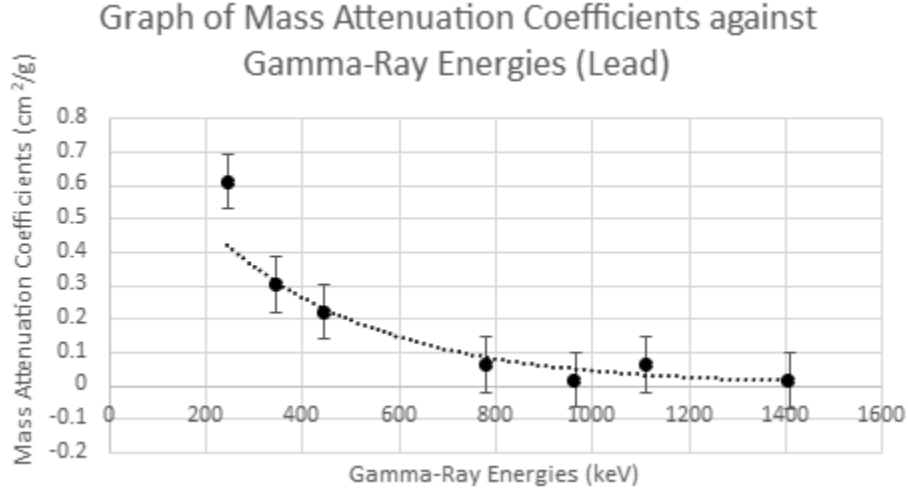


Figure 2: Graph of MAC energy dependence.

MAC values generally decrease following the exponential curve. The only exception is the $1112.18\text{ keV } \mu$ value. Below is a table comparing measured and literature μ values for the 7 gamma-ray energies.

Energy (keV)	Measured Value (cm^2/g)	Literature Value (cm^2/g)	Avg % deviation
244.79	0.612 ± 0.02	0.701	12.7
344.39	0.303 ± 0.01	0.318	4.7
444.1	0.223 ± 0.06	0.197	13.2
779.07	0.065 ± 0.01	0.089	27.0
964.17	0.018 ± 0.3	0.071	74.6
1112.18	0.065 ± 0.1	0.064	1.56
1408.01	0.015 ± 0.09	0.052	71.2

Table 1: Table of measured and literature μ values. For reference values, see NIST XCOM database[1]

A similar relationship as seen in Figure 2 was expected to be seen for Aluminium [1], however, unfortunately the collected data produced a graph with no obvious relationship between μ values and energies (r value fell extremely close to 0). One possible reason for this is the shifting of ROIs during measurement. It is also worth noting that the software used to take measurements of counts (MEASTRO) crashed during the experiment. This could have affected the recorded values for Aluminium.

The half-lives of Copper and Indium were calculated to be 3.20 ± 0.34 mins and 95.9 ± 48.3 mins (≈ 1.6 hours) respectively. Compared to the literature values of 5.12 mins for Copper[2] and 1.7 hours for Indium[3], the relative average deviations in both values are 37.5% and 6.00% respectively.

Indium count rates were measured at energies of 417 keV (Peak 1), 1097 keV (Peak 2) and 1294 keV (Peak 3) over the course of an hour. The collected data was used to construct graphs of $\ln(\Delta N/\Delta t) - t$ for all three peaks. The slope of these graphs were used together with Eq(3) to find three half-life values. A weighted average of these values was then calculated to find a mean half-life value with weightings being proportional to the error in each peak's inferred half-life. Below is the graph constructed for Peak 1.

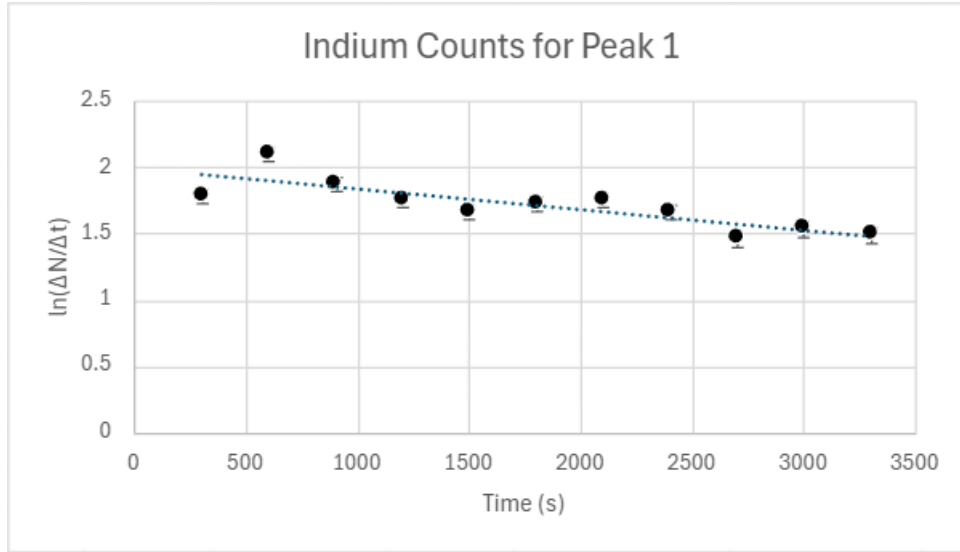


Figure 3: Graph of count rate natural logarithm evolution for gamma-ray energy Peak 1: 417keV. Three such graphs were produced each showing very similar decaying linear relationships.

Copper half-life was calculated by finding the best-fit values to the non-linear least squares problem. A graph of count rate against time was produced.

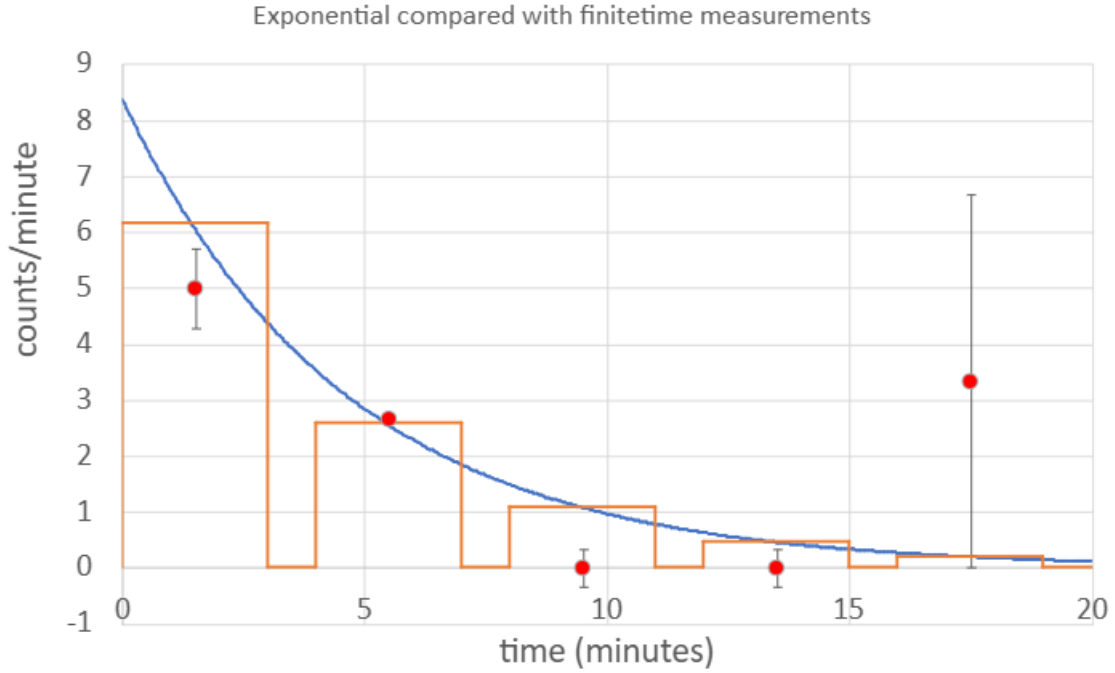


Figure 4: Graph of count rate against time. A decaying exponential relationship is seen from which half-life can be inferred.

4 Discussion

Lead Mass attenuation coefficients for energies of 244.79 keV, 344.79 keV, 444.1 keV and 1112.18 keV were found to be in very close agreement with currently accepted values. Other μ values seemed to deviate significantly from literature. However, overall, the general dependence of MAC values on gamma-ray energies was found as predicted by theory. MAC values were found to decrease exponentially with increasing gamma-ray energies between 244.79 keV and 1408.01 keV.

Data collected for Aluminium didn't show any statistically identifiable relation between these two variables. As previously mentioned, this could be attributed to MAESTRO crashing during the experiment. It is also likely that the high error in measurements was caused by the downward shifting of ROIs over time due to other external factors present in the laboratory (for example unideal temperatures). To counteract this, the ROIs could have been reset between measurements.

Half-lives of Copper and Indium were both measured with the Indium half-life being in very close agreement with literature. The half-life of Copper deviated by 37.5% from literature. This could be from count measurements of the third and fourth acquire periods, both of which measured at 0. These measurement windows yielded such values due to the method used by MEASTRO to account for background radiation.

5 Conclusion

Values of mass attenuation coefficients were obtained for 7 gamma-ray energies in Lead. They were shown to have a decaying exponential dependence on gamma-ray energy as predicted by theory. Half-lives of Copper and Indium were also calculated. The Indium half-life inferred from collected data was in really close agreement with the literature, however, the Copper half-life deviated significantly from the currently expected result.

6 Acknowledgements

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